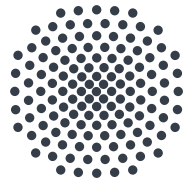
Analysis and Synthesis of Switched Optimization Algorithms

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Carsten Scherer
Fabian Jakob
Andrea Iannelli

ThC01.5

August 27, 2026



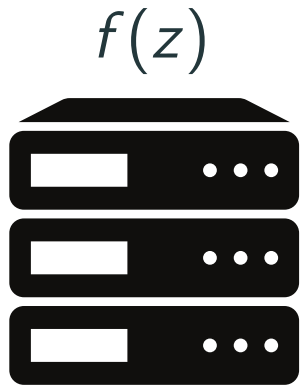
Universität Stuttgart

mst ist

SimTech

Optimization Setup

Want to solve an optimization problem $z^* \in \operatorname{argmin} f(z)$



Examples: Motion planning, regression, filtering

Optimization Setup

Want to solve an optimization problem $z^* \in \operatorname{argmin} f(z)$

Issue: f is only accessible over a **network**



Examples: teleoperation, satellite control, data assimilation

Even Worse

The communication network may have dynamics



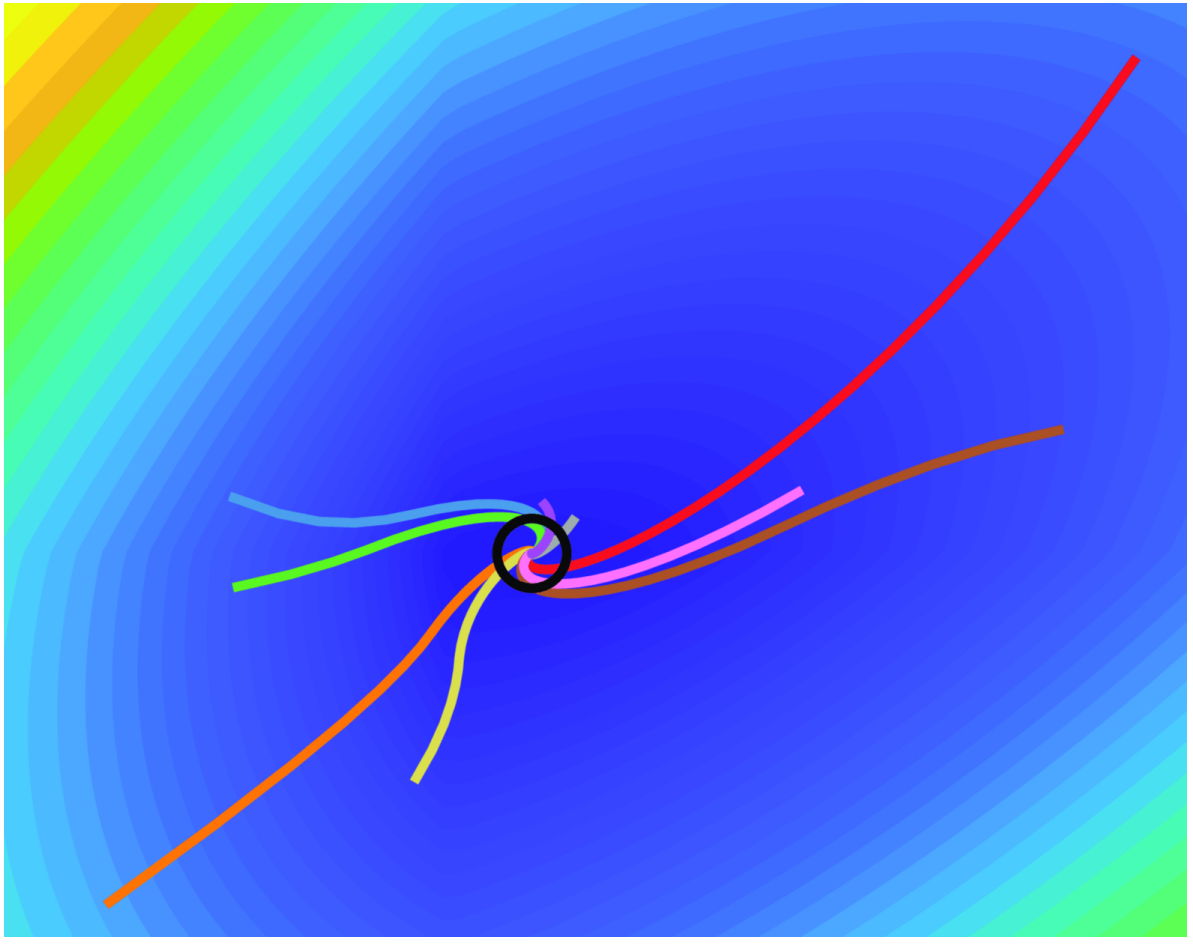
- **Packet Drops**
- Time delays
- Instability
- Channel memory

Still want to optimize in this setting



Algorithm Definition (Simple)

Procedure to generate $\{z_k\}_{k\in\mathbb{N}}$, convergent if $z_k \to z^*$



The Goal

Algorithm should converge as $\lim_{k\to\infty} z_k = z^*$ for all

1. initial iterates z_0
2. functions f in a given class ($S_{m,L}$: m-convex, L-smooth)
3. switching sequences in the network

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3. switching sequences in the network

Should converge *quickly* (exponential convergence)

Main Tasks	
Analysis: check this.	Synthesis: enforce this.

Representations of Algorithms



Algorithms as Dynamical Systems

Gradient descent scheme $z_{k+1} = z_k - \gamma \nabla f(z_k)$

[1]Wang, Jing, and Nicola Elia. "A control perspective for centralized and distributed convex optimization." 2011 50th IEEE conference on decision and control and European control conference. IEEE, 2011.

Algorithms as Dynamical Systems

Gradient descent scheme $z_{k+1} = z_k - \gamma \nabla f(z_k)$

Representation as Luré interconnection [1]

$$w_k = \nabla f(z_k)$$

$$\left(\begin{array}{c} x_{k+1} \\ \hline z_k \end{array}\right) = \left(\begin{array}{c|c} I & -\gamma I \\ \hline I & 0 \end{array}\right) \left(\begin{array}{c} x_k \\ \hline w_k \end{array}\right)$$

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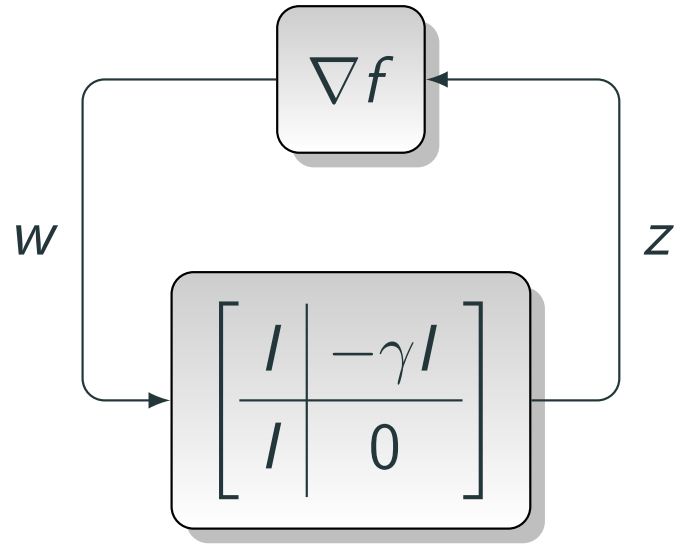
Algorithms as Dynamical Systems

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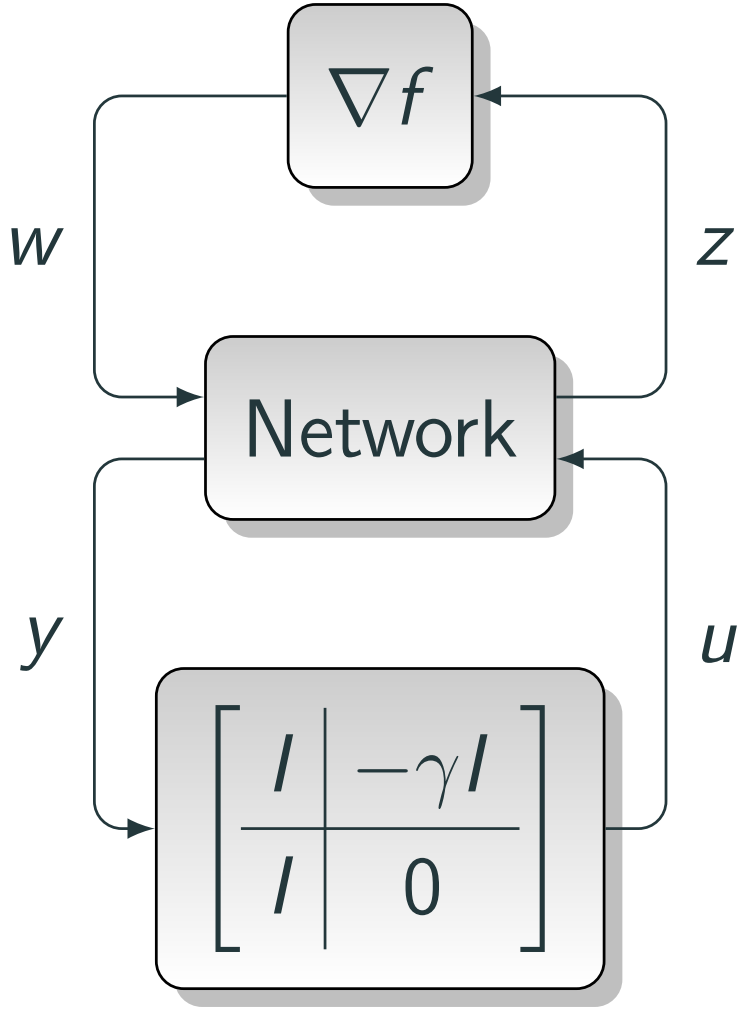
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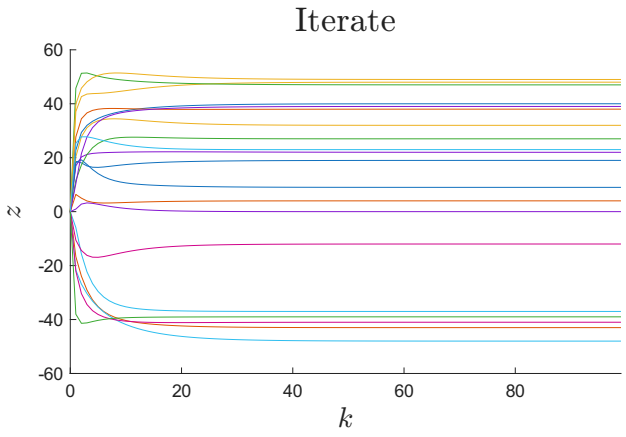
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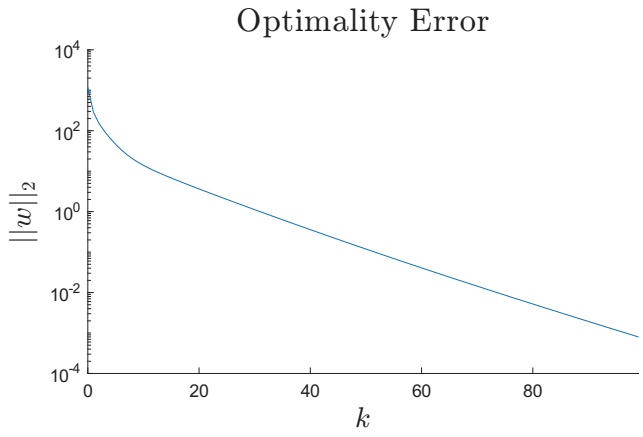
Networked Algorithms

Gradient may not be accessible (e.g. time-delays, cross-talk)

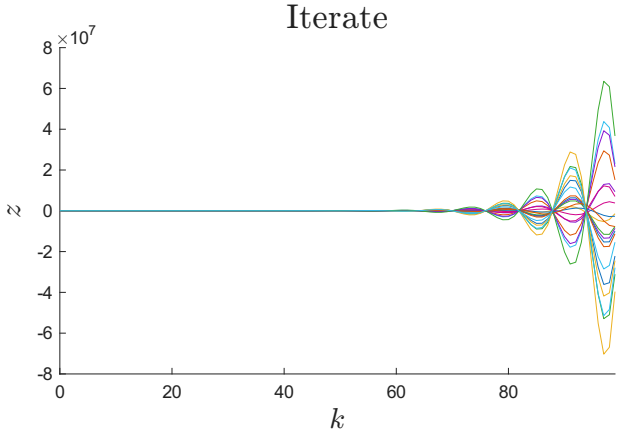


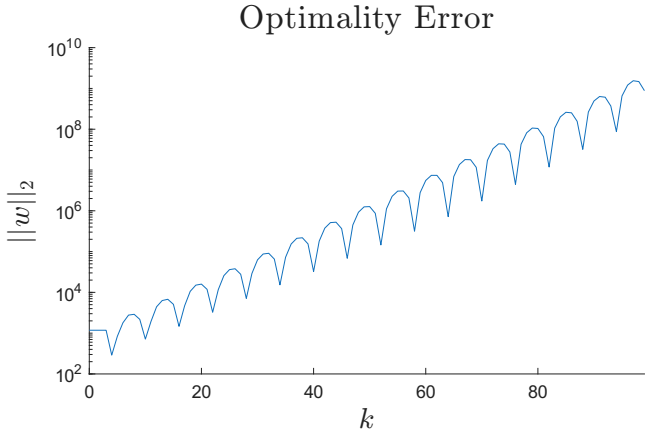
Gradient Descent Example





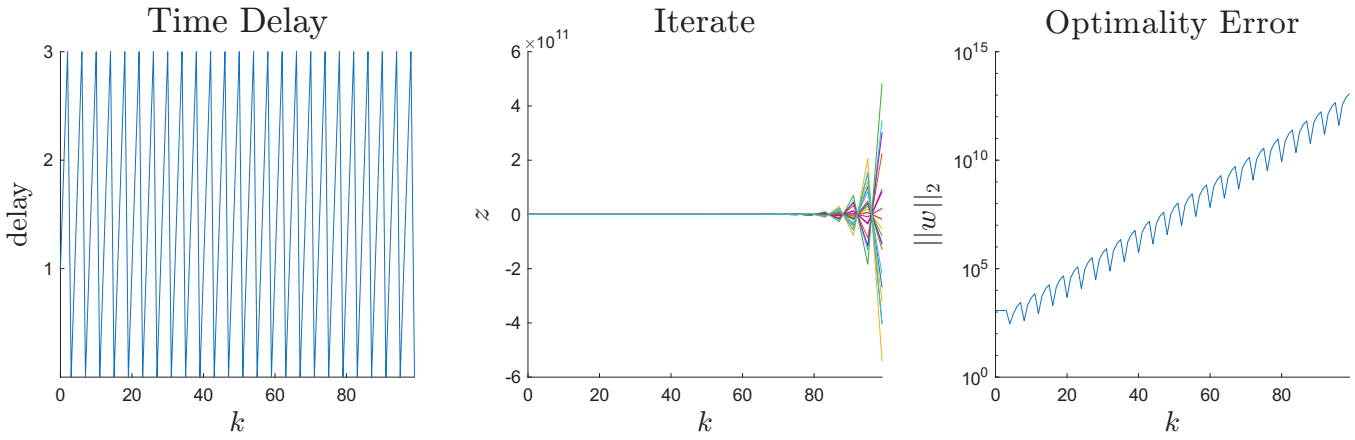
No delay, convergent



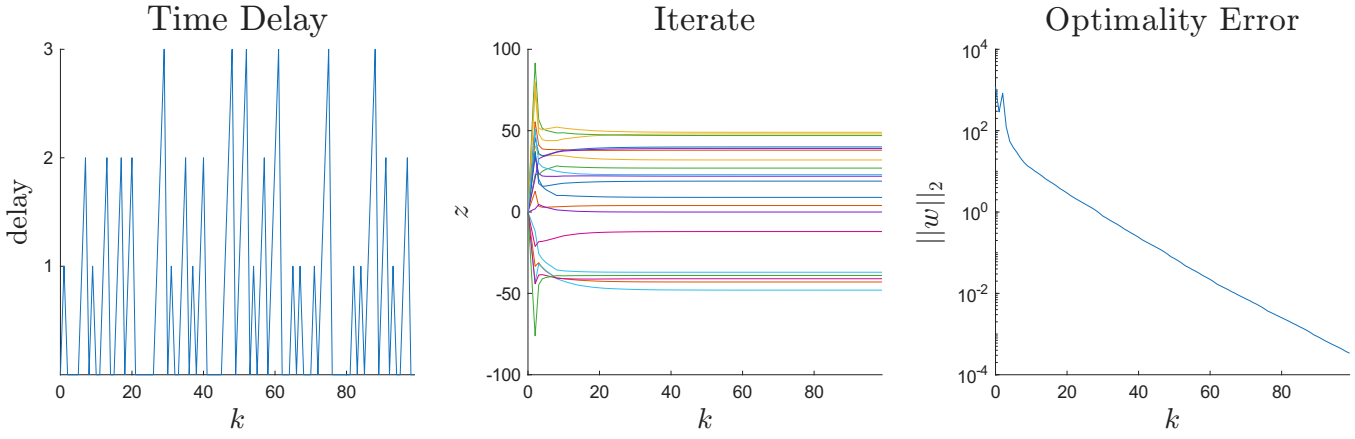


3-step time delay, divergent

Switched Gradient Descent Example



Periodic delay, divergent



Packet drop, convergent (in this specific instance)

Switched Algorithm Modeling

Switched systems with modes $r \in \{1, \ldots, N_s\}$

$$\left(\begin{array}{c} x_{k+1} \\ \hline z_k \end{array}\right) = \left(\begin{array}{c|c} \mathcal{A}_r & \mathcal{B}_r \\ \hline \mathcal{C}_r & 0 \end{array}\right) \left(\begin{array}{c} x_k \\ \hline w_k \end{array}\right), \qquad w_k = \nabla f(z_k)$$

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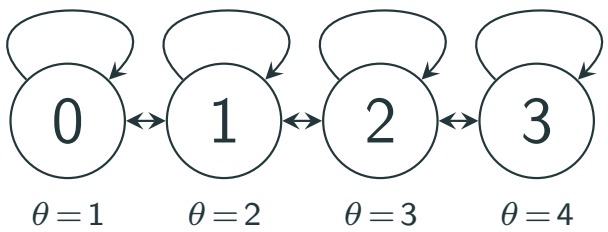
Gradient descent with time-varying delay $h(k)$

$$\text{Rule :} \qquad z_{k+1} = z_k - \nabla f(z_{k-h(k)})$$

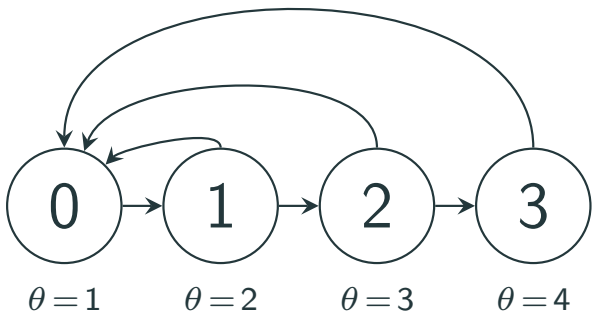
$$\underbrace{\left(\begin{array}{ccc|c} 0 & 0 & I & I \\ I & 0 & 0 & 0 \\ 0 & 0 & I & -\gamma I \\ \hline 0 & 0 & I & 0 \end{array}\right)}_{h=0,\theta=1}, \quad \underbrace{\left(\begin{array}{ccc|c} 0 & 0 & I & I \\ I & 0 & 0 & 0 \\ 0 & 0 & I & -\gamma I \\ \hline I & 0 & 0 & 0 \end{array}\right)}_{h=1,\theta=2}, \quad \underbrace{\left(\begin{array}{ccc|c} 0 & 0 & I & I \\ I & 0 & 0 & 0 \\ 0 & 0 & I & -\gamma I \\ \hline 0 & I & 0 & 0 \end{array}\right)}_{h=2,\theta=3}$$

Transition Graph

Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of allowable transitions



Bounded rate delay



Packet drop

Sequence $(\theta_0, \theta_1, \theta_2, \ldots)$ restricted by $(\theta_k, \theta_{k+1}) \in \mathcal{E}$ always

Convergence Properties

Algorithm is run until a fixed orbit $(\{x_r^*\}, w^*, z^*)$ is reached

$$\left(\begin{array}{c} x_{r'}^* \\ \hline z^* \end{array}\right) = \left(\begin{array}{c|c} \mathcal{A}_r & \mathcal{B}_r \\ \hline \mathcal{C}_r & 0 \end{array}\right) \left(\begin{array}{c} x_r^* \\ \hline w^* \end{array}\right), \qquad w^* \in \nabla f(z^*)$$

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Convergent at rate $\rho \in (0, 1)$ if fixed orbit exists with

Zero-State: $z^* = 0$ implies $x_r^* = 0$ for all modes r

Optimality: $w^* = 0$

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Zero-State: $z^* = 0$ implies $x_r^* = 0$ for all modes r

Optimality: $w^* = 0$

ρ-Attractivity: $\forall x_0, \theta : \lim_{k\to\infty} \begin{pmatrix} x_k - x_{\theta_k^*} \\ z_k - z^* \\ w_k \end{pmatrix} \rho^{-k} \to 0$

Association to Linear Switched Systems

Optimization of quadratic $f(z) = \frac{m}{2}\|z - z^*\|_2$ inside $S_{m,L}$

$$\left(\begin{array}{c} x_{k+1} \\ \hline z_k - z^* \end{array}\right) = \left(\begin{array}{c|c} \mathcal{A}_{\theta_k} + \mathcal{B}_{\theta_k} m \mathcal{C}_{\theta_k} & -\mathcal{B}_{\theta_k} m \\ \hline \mathcal{C}_{\theta_k} & -I \end{array}\right) \left(\begin{array}{c} x_k \\ \hline z^* \end{array}\right),$$

Algorithm convergence if $z - z^* \to 0$ for all z^*

This is output regulation of linear switched systems [2]

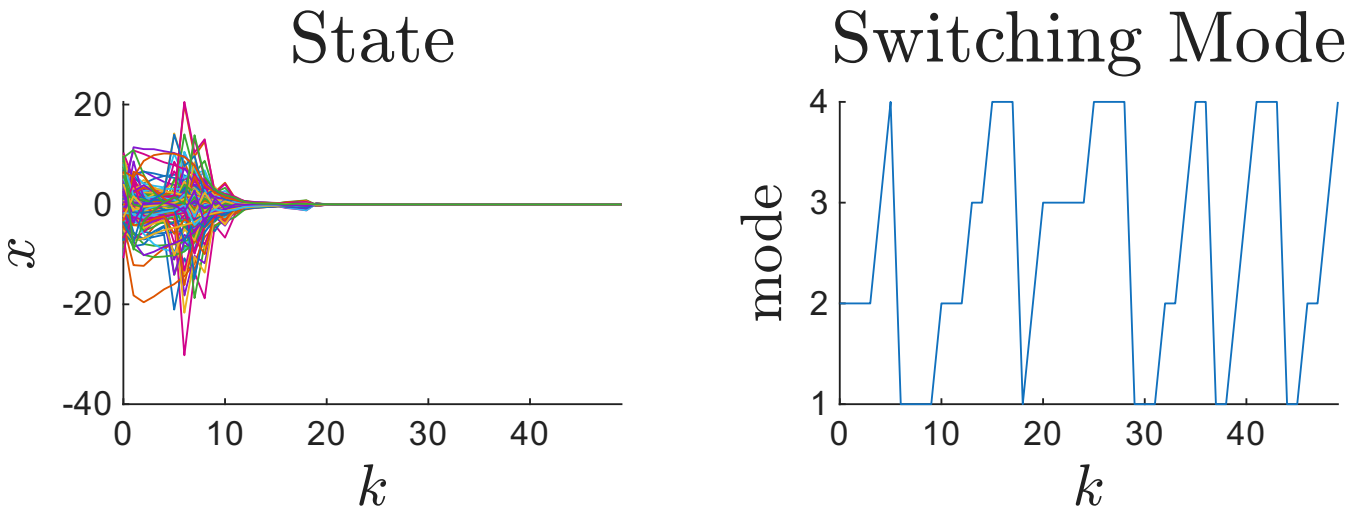
[2] Zattoni, Elena, Anna Maria Perdon, and Giuseppe Conte. "The output regulation problem with stability for linear switching systems: A geometric approach." Automatica 49.10 (2013): 2953-2962.

Convergence 1: Robust Stability

Robust ρ-Stability: If $z^* = 0$, then $\lim_{k\to\infty} \rho^{-k} x_k = 0$

$$x_{k+1} = (\rho^{-1}\mathcal{A}_{\theta_k})x_k + (\rho^{-1}\mathcal{B}_{\theta_k})w_k, \qquad w_k = \nabla f(z_k),$$
$$z_k = \mathcal{C}_{\theta_k} x_k.$$

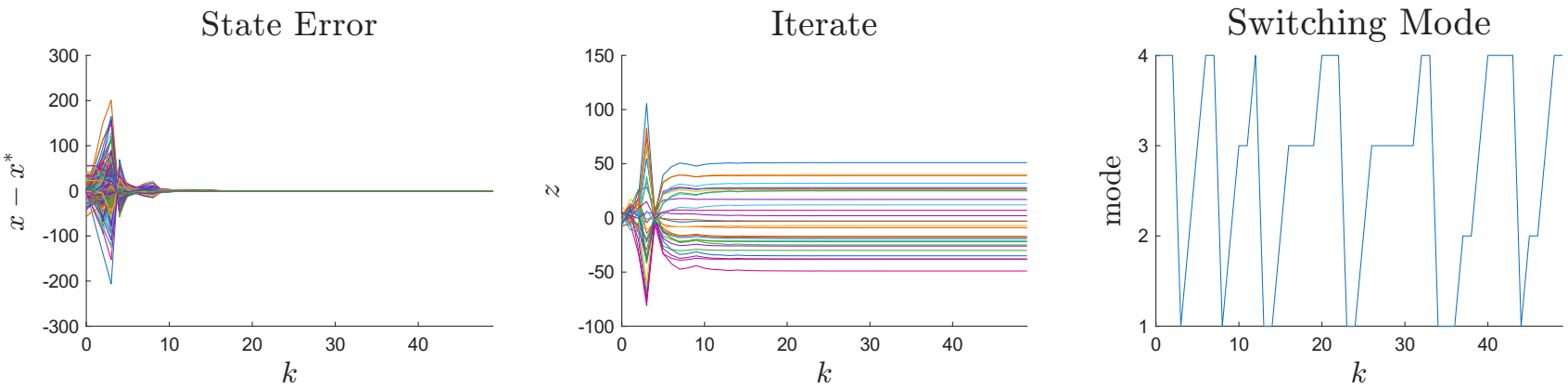
for all initial conditions x_0 and sequences θ



Convergence 2: Regulator Equation

Regulator Equations: Exists solution to linear equations

$$\forall (r, r') \in \mathcal{E}: \qquad \mathcal{A}_r \mathbf{X}_r - \mathbf{X}_{r'} = 0, \qquad \mathcal{C}_r \mathbf{X}_r = I$$



Globally attractive fixed orbit $(\mathbf{X}_r z^*, 0, z^*)$

Analysis

LMI Test for Stability

Sequences $w \in \nabla f(z)$ satisfy IQC/dissipation relations

Assemble system from $(\mathcal{A}_r, \mathcal{B}_r, \mathcal{C}_r)$, class (m, L), parameters $\boldsymbol{\lambda}$

$$[\star]^\top \begin{pmatrix} M_{r'} & 0 \\ 0 & -M_r \end{pmatrix} \begin{pmatrix} \hat{\mathcal{A}}_r^{\boldsymbol{\lambda}} & \hat{\mathcal{B}}_r^{\boldsymbol{\lambda}} \\ I & 0 \end{pmatrix} + [\star]^\top \begin{pmatrix} 0 & I_c \\ I_c & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathcal{C}}_r^{\boldsymbol{\lambda}} & \hat{\mathcal{D}}_r^{\boldsymbol{\lambda}} \\ 0 & I_c \end{pmatrix} \prec 0.$$

LMI Test for Stability

Sequences $w \in \nabla f(z)$ satisfy IQC/dissipation relations

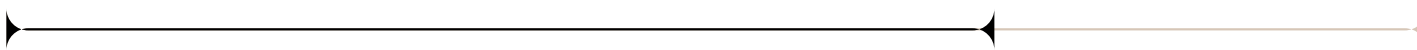
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Feasibility of LMI $\forall (r, r') \in \mathcal{E}$ proves ρ-exponential stability

$$\|x_k - x^*_{\theta_k}\|_2 \leq \gamma \rho^k \|x_k - x^*_{\theta_0}\|_2$$

Synthesis



Networked Algorithm Design

System $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ arises from interconnection

Network:
$$\left(\begin{array}{c} v_{k+1} \\ \hline z_k \\ y_k \end{array}\right) = \left(\begin{array}{c|cc} A_r & B_{r,1} & B_{r,2} \\ \hline C_{r,1} & D_{r,11} & D_{r,12} \\ C_{r,2} & D_{r,21} & D_{r,22} \end{array}\right) \left(\begin{array}{c} v_k \\ \hline w_k \\ u_k \end{array}\right),$$

Controller:
$$\left(\begin{array}{c} \xi_{k+1} \\ \hline u_k \end{array}\right) = \left(\begin{array}{c|c} A_r^K & B_r^K \\ \hline C_r^K & D_r^K \end{array}\right) \left(\begin{array}{c} \xi_k \\ \hline y_k \end{array}\right)$$

Synthesis:

Design controllers ensuring ρ-convergence

Regulation-Based Design

Closed loop system must satisfy convergence criteria

Regulation-Based Design

Closed loop system must satisfy convergence criteria

Robust Stability: Switched robust control design (LMI)

Regulation-Based Design

Closed loop system must satisfy convergence criteria

Robust Stability: Switched robust control design (LMI)

Regulator Equations: Exists solution $(\Pi_r, \Gamma_r, \Phi_r)$ to

$$\forall (r, r') \in \mathcal{E} : \quad \left(\begin{array}{c|cc} A_r & B_{r,1} & B_{r,2} \\ \hline C_{r,1} & D_{r,11} & D_{r,12} \\ C_{r,2} & D_{r,21} & D_{r,22} \end{array}\right) \begin{pmatrix} \Pi_r \\ I \\ \Gamma_r \end{pmatrix} = \begin{pmatrix} \Pi_{r'} \\ 0 \\ \Phi_r \end{pmatrix}$$

Regulation-Based Design

Closed loop system must satisfy convergence criteria

Robust Stability: Switched robust control design (LMI)

Regulator Equations: Exists solution $(\Pi_r, \Gamma_r, \Phi_r)$ to

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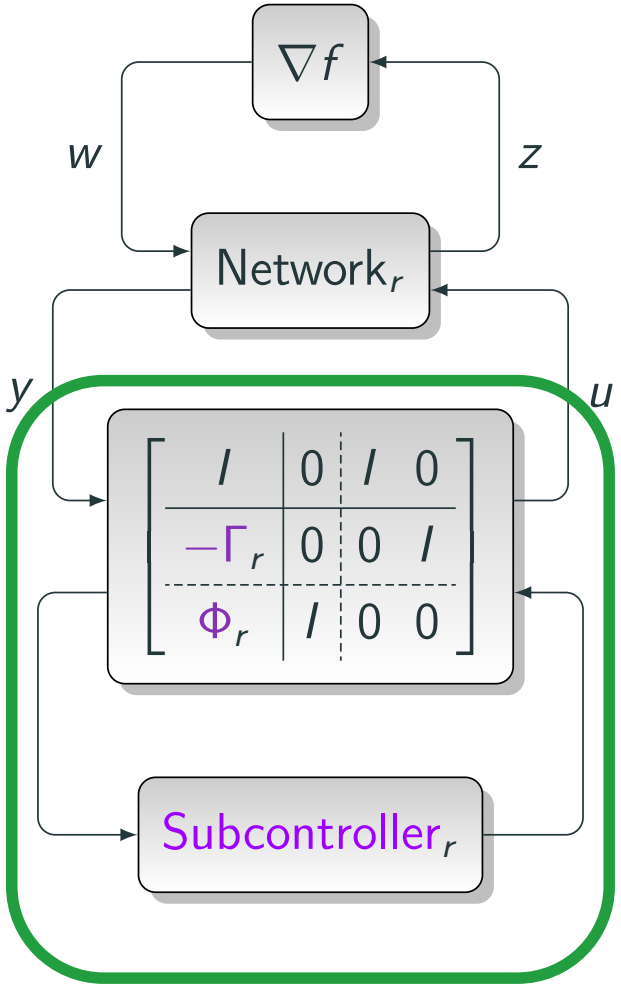
Fixed orbits have $(v_r^*, u_r^*, y^*) = (\Pi_r z^*, \Gamma_r z^*, \Phi_r z^*)$

Requires new assumption: $I - D_{r,11} m$ invertible

Full-Order Internal Models and Synthesis

Add a model in the middle

Regulator Equation *always* satisfied

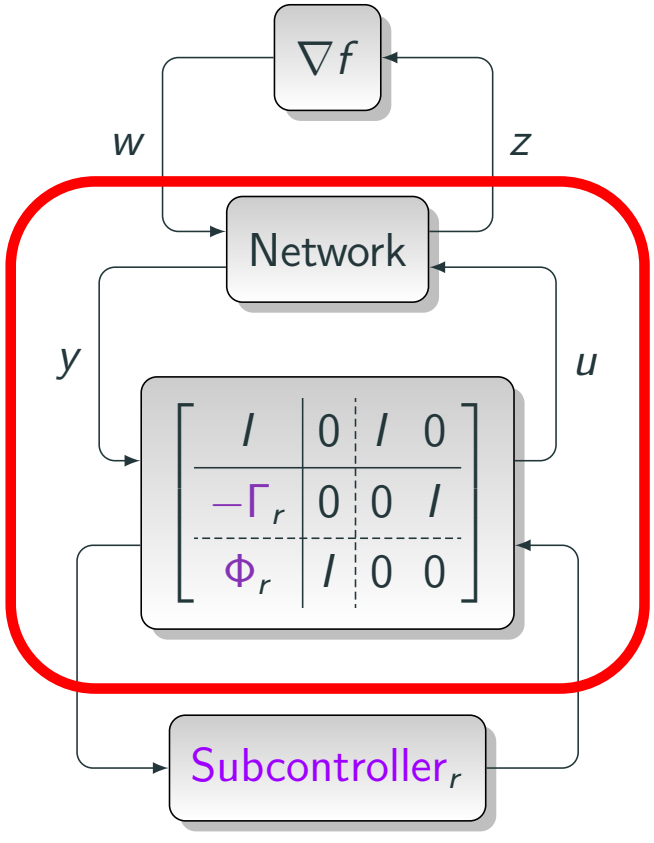


Full-Order Internal Models and Synthesis (cont.)

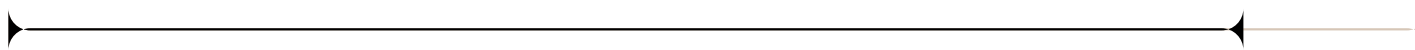
Accumulate Network and Model

Robust-control design of
Subcontroller

Allows for performance
(e.g. convergence rates, tracking)

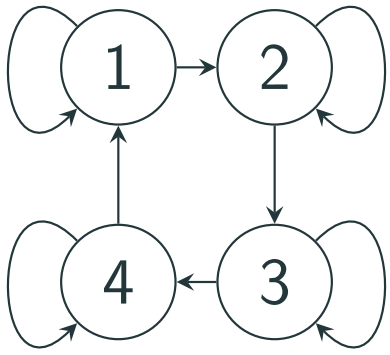


Profile Example



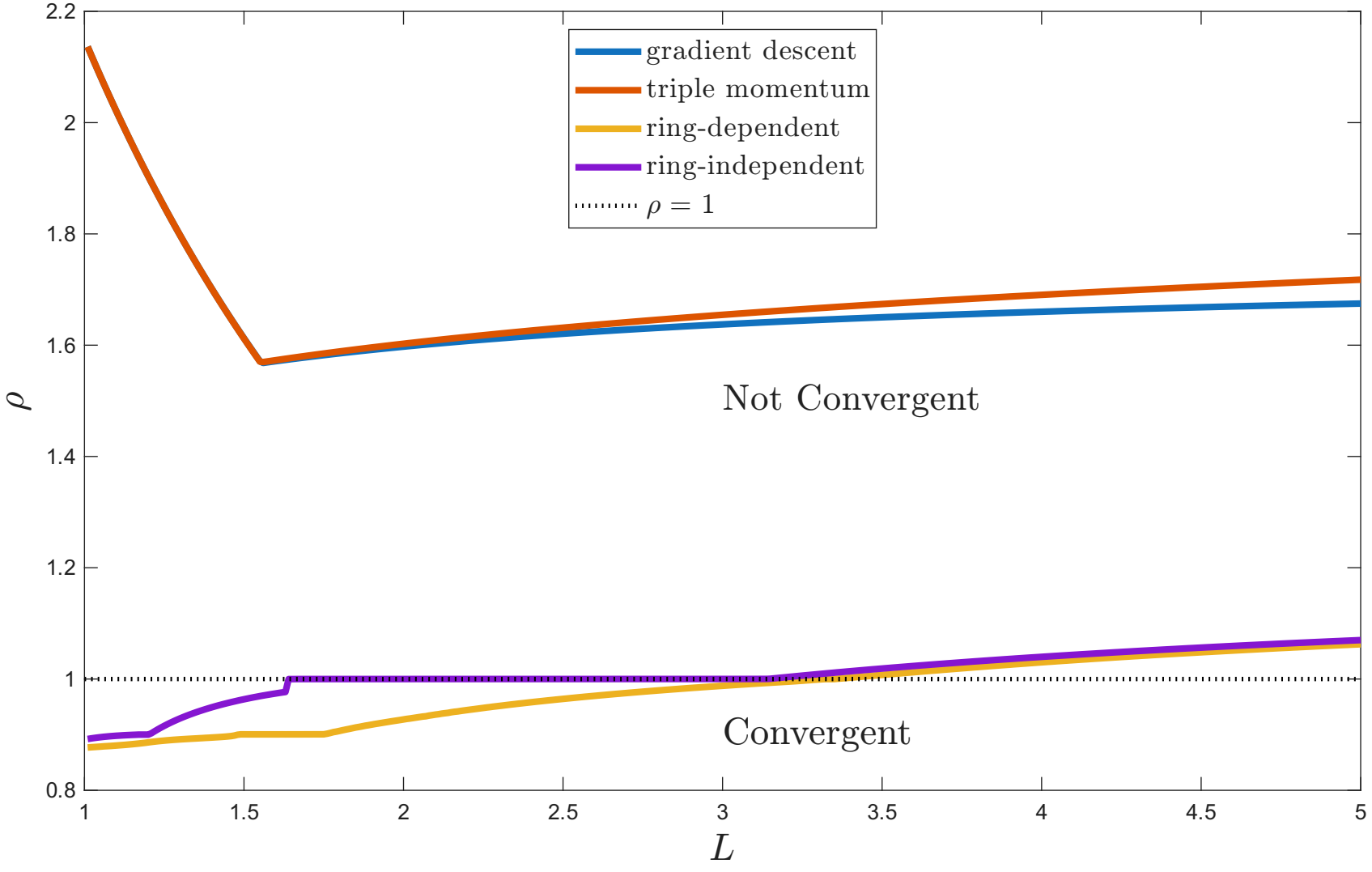
Ring Network

Ring-structured switching graph



Second order dynamics, mode $\theta = 4$ is unstable

Ring Network Sweep



Profile against nominal Gradient Descent, Triple Momentum

Wrap-Up

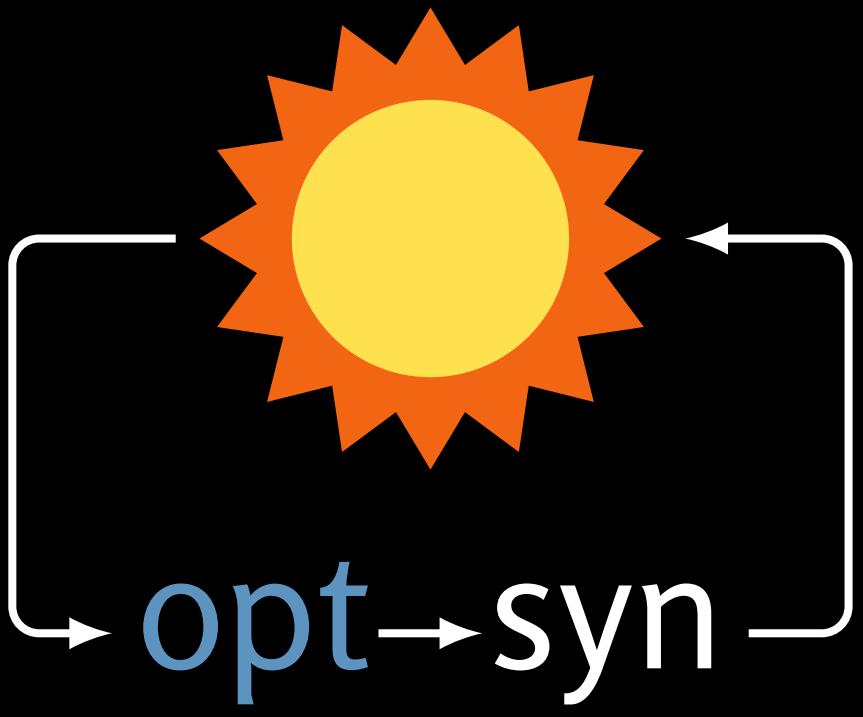
Switched network dynamics can break optimization algorithms

Sufficient convergence criteria:

- Robust Stability
- Regulator Equation

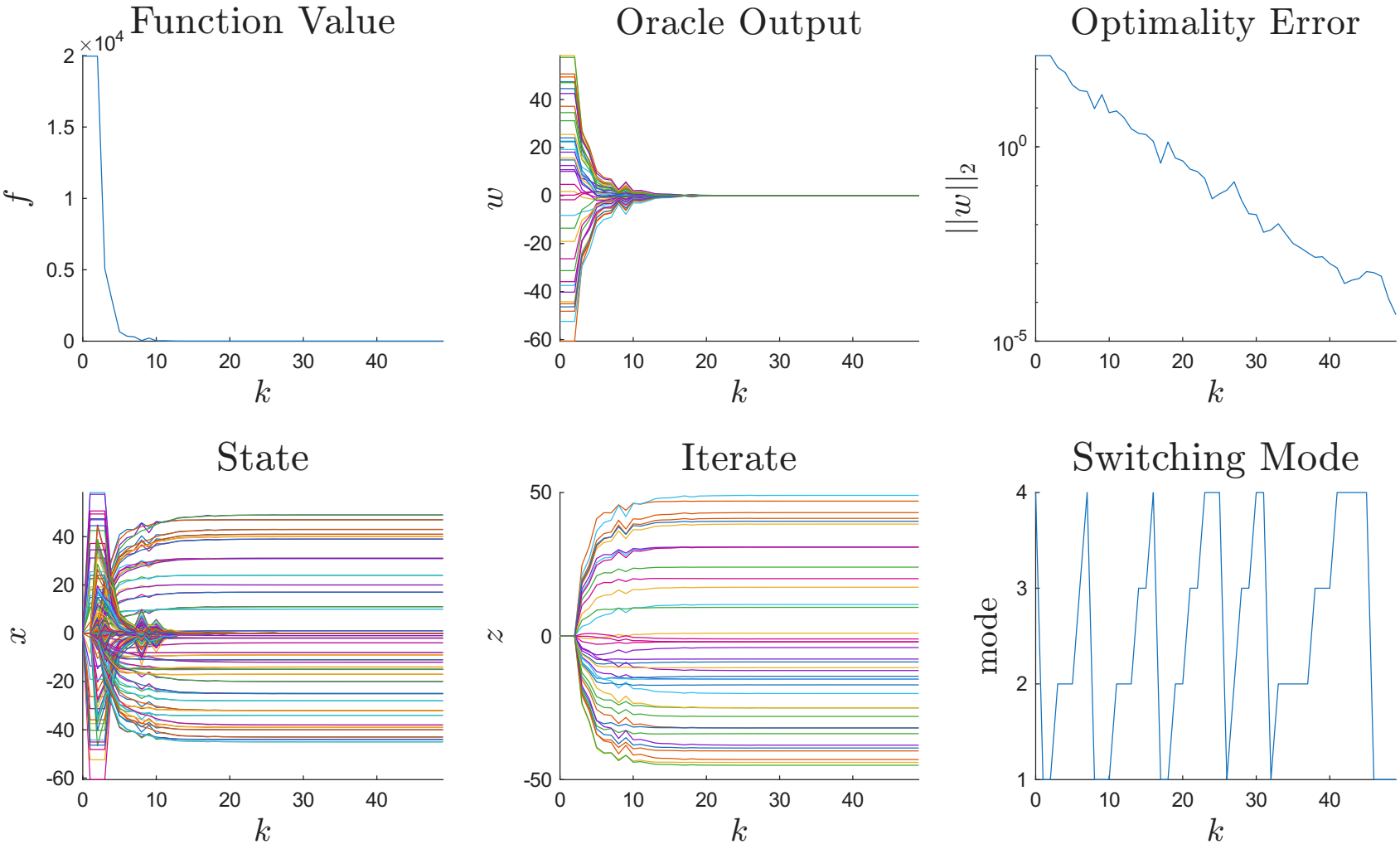
Certified switched algorithm analysis/design by solving LMIs

Optimize(d) under Switches!



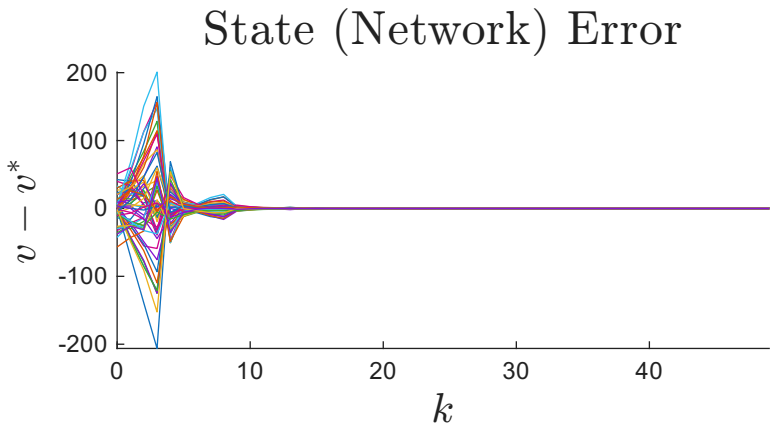
Bonus Slides

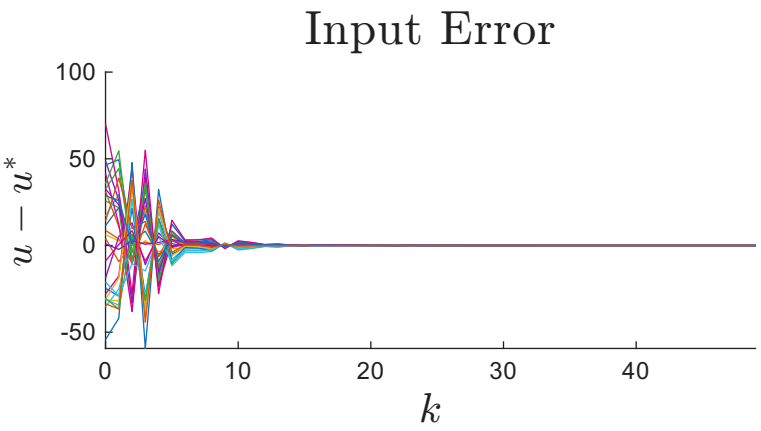
Ring Network Execution

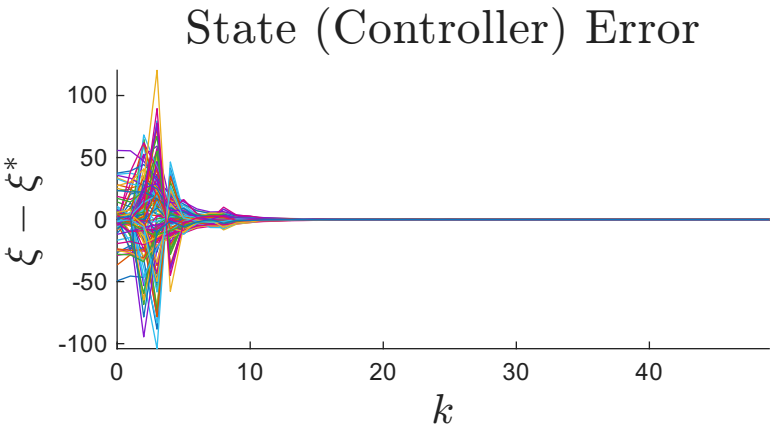


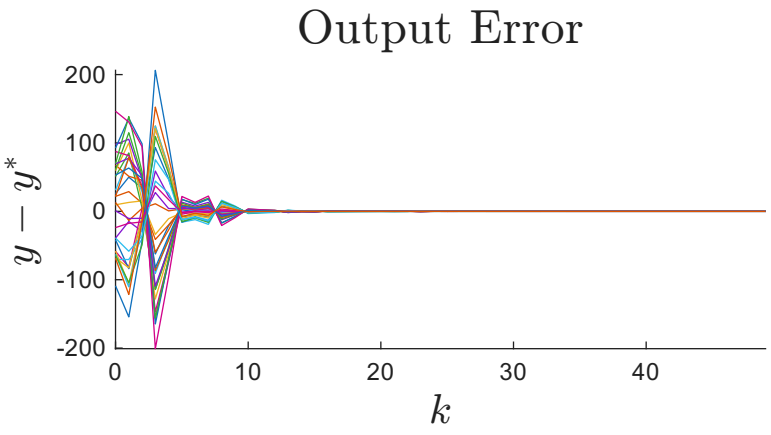
Convergence with unstable network dynamics, $m = 1, L = 1.5$

Ring Network Tracking









Signals track fixed orbit, based on Regulator Equation

Valid Relations

Subgradient inequality $f \in \mathcal{S}_{0,\infty}$, $(x, x') \in \mathrm{dom}(f)$, $g \in \partial f(x)$

$$f(x') \geq f(x) + g^\top (x' - x). \tag{2}$$

Can be generalized to $\mathcal{S}_{m,L}$, ρ-weightings

Define $\mathbf{q} = \frac{1}{2}\|\cdot\|_2^2$ $V(x, g) := f(x) - m\mathbf{q}(x) - \sigma\mathbf{q}(g - mx)$

Given $f \in \mathcal{S}_{m,L}$, $\alpha_i > 0$, $\bar{x}_i \in \mathbb{R}^n$, $\bar{g}_i \in \alpha_i^{-1}\partial f(\alpha_i \bar{x}_i)$:

$$V(\alpha_1\bar{x}_1, \alpha_1\bar{g}_1) - V(\alpha_2\bar{x}_2, \alpha_2\bar{g}_2) \leq$$
$$\alpha_1(\bar{g}_1 - m\bar{x}_1)^\top [\alpha_1(\sigma L\bar{x}_1 - \sigma g_1) - \alpha_2(\sigma L\bar{x}_2 - \sigma\bar{g}_2)], \tag{3}$$

Relations satisfied by Gradients

Specific class of functions $f \in S_{m,L}$ with $f(0) = 0, \nabla f(0) = 0$

From sequences (w, z) with $w_k = \nabla f(z_k)$, derive

$$q_k := w_k - mz_k, \qquad p_k := Lz_k - w_k$$

[3] Extends to ρ-discounting: Scherer, Carsten W. "Robust exponential stability and invariance guarantees with general dynamic O'Shea-Zames-Falb multipliers." IFAC-PapersOnLine 56.2 (2023): 5799-5804.

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(p, q) satisfy passivity relations

$$\sum_{k=0}^{T-1} q_k^\top \left(\sum_{\nu=0}^{\nu_{\max}} \boldsymbol{\lambda}_\nu p_k \right) \geq 0 \qquad \forall\ T \in \mathbb{N}, T > 0.$$

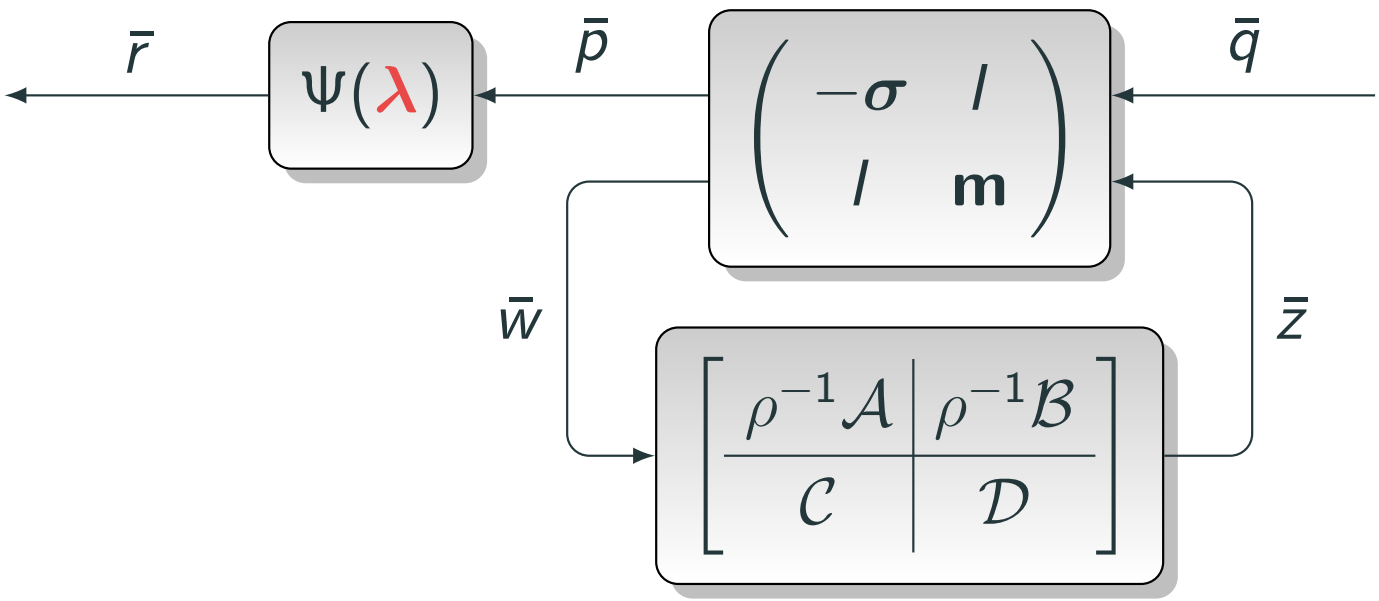
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Filtering Framework

One-step delay operator $\mathbf{z}$, filter coefficients λ^i

Zames-Falb filter: $\Psi(\boldsymbol{\lambda}) = \mathrm{blkdiag}(\bar{p}_t^i \to \sum_{\nu=0}^{\nu_{\max}} \lambda_\nu^i \bar{p}_{t-\nu}^i) \otimes I_c$

$$\sum_{\nu=0}^{\nu_{\max}} \rho^{-\nu} \lambda_\nu > 0 \quad \forall \nu \geq 1: \ \lambda_\nu \leq 0 \tag{4}$$



Zames-Falb Filters

Conic combinations of dissipation relations $\{\mu_\nu\}_{\nu=0}^{\nu_{\max}} \geq 0$

$$\sum_{k=0}^{T-1} \left(\mu_0 \bar{q}_k^\top \bar{p}_k + \sum_{\nu=1}^{\nu_{\max}} \mu_r \bar{q}_k^\top (\bar{p}_k - \rho^\nu \bar{p}_{k-\nu}) \right). \tag{5}$$

Summarized by λ coefficients

$$\sum_{\nu=0}^{\nu_{\max}} \rho^{-\nu} \lambda_\nu > 0 \quad \forall \nu \geq 1 : \ \lambda_\nu \leq 0 \tag{6}$$

Association between λ and ν

$$\lambda_0 = \sum_{\nu=0}^{\nu_{\max}} \mu_n u \qquad \lambda_\nu = -\rho^\nu \mu_\nu \tag{7}$$